IMY 220

Practical 1

Docker Fundamentals

Due: Friday 24 July @ 08:20

The submission instructions are available on ClickUP. Any deviation from these instructions will result in a 10% deduction from your mark.

PhotoShare Development Sprint

You have joined the development team responsible for the **PhotoShare** application. The backend API has already been developed, but before it can be deployed it needs to be containerised using Docker.

Your task is to containerise the supplied Node.js and Express application so that it can be run consistently on any machine using Docker.

Instructions

- The aim of this practical is to demonstrate your understanding of the Docker concepts covered during the recap lecture by containerising a supplied Node.js and Express application.
- Download **practical-1.zip** from ClickUP.
- The supplied project contains a working PhotoShare API. You are **not** required to modify the functionality of the application.
- You are required to Dockerise the application according to the requirements below.
- Include a text file called **instructions.txt** containing all Docker commands required to build and run your Docker image.

Section 1

Dockerise the supplied Express application by completing the following tasks:

- Create a **Dockerfile** for the application.
- Use the **node:24-alpine** Docker image as the base image.
- Set an appropriate working directory inside the container.
- Ensure that unnecessary files (such as **node_modules**) are not copied into the Docker image.
- Copy the required application files into the container.
- Install the application's dependencies.

- Document the application's listening port using **EXPOSE**.
- Configure the container so that it starts the application automatically.

Section 2

Build and run the Docker image according to the following requirements:

- Name your Docker image **uXXXXXXXX**, where **XXXXXXXX** is your student number.
- Map **host port 3001** to **container port 5000**.
- Supply the following environment variables when running the container:

Environment Variable	Value
APP_NAME	PhotoShare API
APP_ENV	docker

After running the container successfully, verify that the following endpoints are accessible:

- `/`
- `/api/posts`
- `/api/status`

Submission Files

Submit **only** the following files:

- `server.js`
- `Dockerfile`
- `.dockerignore`
- `instructions.txt`

Submission Instructions

- Compress all submission files into a ZIP archive called:

`p1-{student_number}.zip`

where `{student_number}` is your student number.

- Submit your ZIP file to ClickUP before **08:20**.